

Section A

Answer at least two questions from this section.

- 1 (a) Draw labelled diagrams to show crystalline structures of
- (i) magnesium oxide,
 - (ii) ice. [2]
- (b) Using the diagrams in (a) where necessary, explain the following observations:
- (i) magnesium oxide is partially soluble in water
 - (ii) melting point of ice is less than that of magnesium oxide
 - (iii) ice floats on water
 - (iv) lattice energy of magnesium oxide is greater than that of sodium chloride which has a similar crystalline structure [4]
- (c) (i) Construct a Born-Haber cycle for the formation of magnesium oxide from its elements.
- (ii) Use the cycle to calculate the enthalpy change of formation of magnesium oxide. Use the following data as well as relevant data from the Data Booklet.
- $\Delta H_{\text{at}}(\text{Mg}) = +150 \text{ kJmol}^{-1}$
 $\Delta H_{\text{lat}}(\text{MgO}) = -3933 \text{ kJmol}^{-1}$
 First electron affinity of oxygen = -142 kJmol^{-1}
 Second electron affinity of oxygen = $+775 \text{ kJmol}^{-1}$
- (iii) Explain why the second electron affinity of oxygen is endothermic. [6]
- [Total: 12]
- 2 (a) Define a *Bronsted-Lowry acid*. [1]
- (b) Explain the following observations:
- (i) $\text{CCl}_3\text{CO}_2\text{H}_{(\text{aq})}$ is a stronger acid than $\text{CH}_3\text{CO}_2\text{H}_{(\text{aq})}$
 - (ii) $[\text{Al}(\text{H}_2\text{O})_6]^{3+}_{(\text{aq})}$ forms a stronger acid than $[\text{Mg}(\text{H}_2\text{O})_6]^{2+}$ [3]

- (c) A 15.0 cm^3 sample of 0.1 mol dm^{-3} aqueous sodium hydroxide was added to 25.0 cm^3 of 0.1 mol dm^{-3} ethanoic acid of pK_a value 4.76.

- (i) Explain, with the aid of equations, why the resulting solution can act as a buffer solution.
- (ii) Calculate the pH of the resulting solution.
- (iii) Draw a sketch of the titration curve obtained when 50.0 cm^3 of 0.1 mol dm^{-3} ethanoic acid is titrated with 25.0 cm^3 of 0.1 mol dm^{-3} sodium hydroxide.

[8]
[Total: 12]

[1]

- 3 (a) Define the term *half life*.

- (b) Hydrolysis of methylethanoate, $\text{CH}_3\text{COOCH}_3$, is catalysed by hydrogen ions, H^+ . The following results were obtained when the concentration of the ester was measured as a function of time, t , at different hydrogen ion concentrations.

At $[\text{H}^+] = 0.3 \text{ mol dm}^{-3}$:

[ester]/ mol dm^{-3}	1.00	0.64	0.41	0.26
t/s	0.00	1500	3000	4500

At $[\text{H}^+] = 0.6 \text{ mol dm}^{-3}$:

[ester]/ mol dm^{-3}	1.00	0.55	0.30	0.17
t/s	0.00	1000	2000	3000

- (i) Write the equation for the acid hydrolysis of methylethanoate.
- (ii) Using the same axes, plot graphs of [ester] against time at the different concentrations of hydrogen ions.
- (iii) Using the initial rates-method, determine from the graphs, the order of the reaction with respect to
- [ester],
 - [hydrogen ions].
- (iv) Write the rate equation for the reaction.

[11]
[Total: 12]

Section B

Answer at least one question from this section.

- 4 Sulphur dioxide is produced by the action of hot concentrated sulphuric acid on copper.

(a) Write the equation for the reaction. [1]

(b) (i) State the observation(s) made when sulphur dioxide is bubbled through an aqueous solution of potassium manganate (VII).

(ii) Write the equation for the reaction in (i) and calculate its E^θ cell.

Use of the Data Booklet is relevant to this question. [3]

(c) One effect of sulphur dioxide on the environment is acid-rain formation.

(i) Give one effect of acid-rain on the environment.

(ii) Explain, by means of equations, how oxides of nitrogen catalyse acid-rain formation. [3]

(d) (i) Write equations to show how sulphuric acid is formed from sulphur trioxide.

(ii) Describe the observations made when concentrated sulphuric acid is added to

1. solid sodium chloride,

2. sodium iodide.

[5]
[Total: 12]

- 5 (a) Compare and contrast the chemistry of bromine and iodine with respect to each of the following reactions. Suggest explanations for any differences in behaviour towards each reagent:

(i) reactions of the elements with hydrogen gas

(ii) reactions of silver bromide and silver iodide with aqueous ammonia [6]

(b) An excess of aqueous potassium iodide was added to a portion of 25 cm^3 of aqueous bromine. The iodine liberated required 19.50 cm^3 of a $0.120 \text{ mol dm}^{-3}$ sodium thiosulphate.

(i) Write equations for the reactions that occurred.

(ii) Calculate the number of moles of iodine liberated.

(iii) A second portion of 25.0 cm^3 of the aqueous bromine required 2.45 cm^3 of the 0.12 mol dm^{-3} sodium thiosulphate and sulphate ions were detected as one of the products.

1. Calculate the number of moles of thiosulphate that reacted with the bromine.

2. Hence derive an equation for the reaction between bromine and thiosulphate ions.

(iv) Explain the differences in the reactions of bromine and iodine with aqueous sodium thiosulphate.

[6]

Section C

Answer at least **two** questions from this section.

6 Fig 1.1 shows the structure of an organic compound, X.

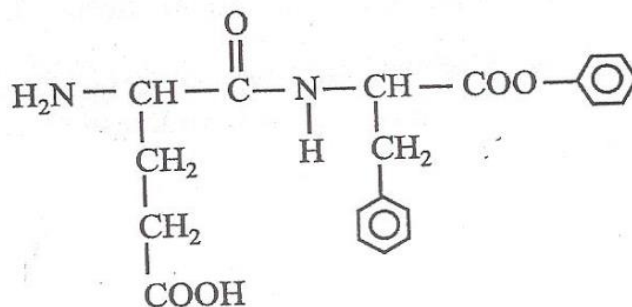


Fig. 1.1

- (a) (i) Name any **four** functional groups in X.
- (ii) Draw the structures of the organic products formed by complete hydrolysis of X using hydrochloric acid.

[5]

(b) Fig. 1.2 shows the structure of glutamic acid.

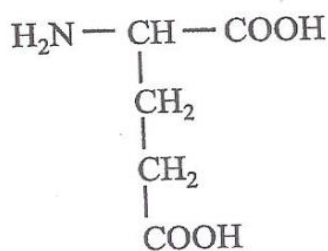


Fig. 1.2

- (i) Explain how glutamic acid reacts with
1. water,
 2. dilute $\text{HCl}_{(\text{aq})}$.
- (ii) Name the type of isomerism shown by glutamic acid.
- (iii) Draw the isomers of glutamic acid.
- (c) Explain why glutamic acid is a solid at room temperature with a relatively high melting point.

[5]

[2]

[Total: 12]

Fig. 2.1 shows the structure of procaine chloride, a chemical derived from nitrogen compounds.

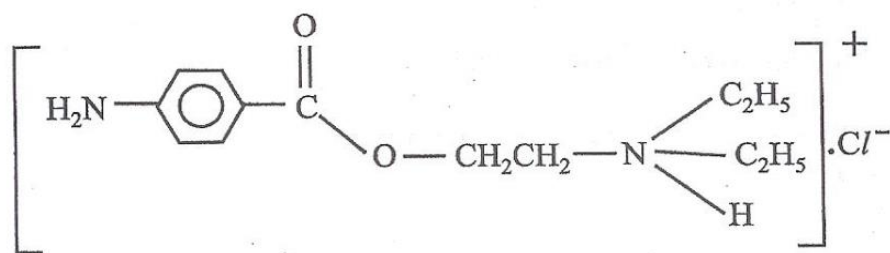


Fig. 2.1

- (a) Suggest, with a reason, the solubility of procaine chloride in water. [2]
- (b) **Fig. 2.2** shows a reaction scheme used in the synthesis of procaine chloride.

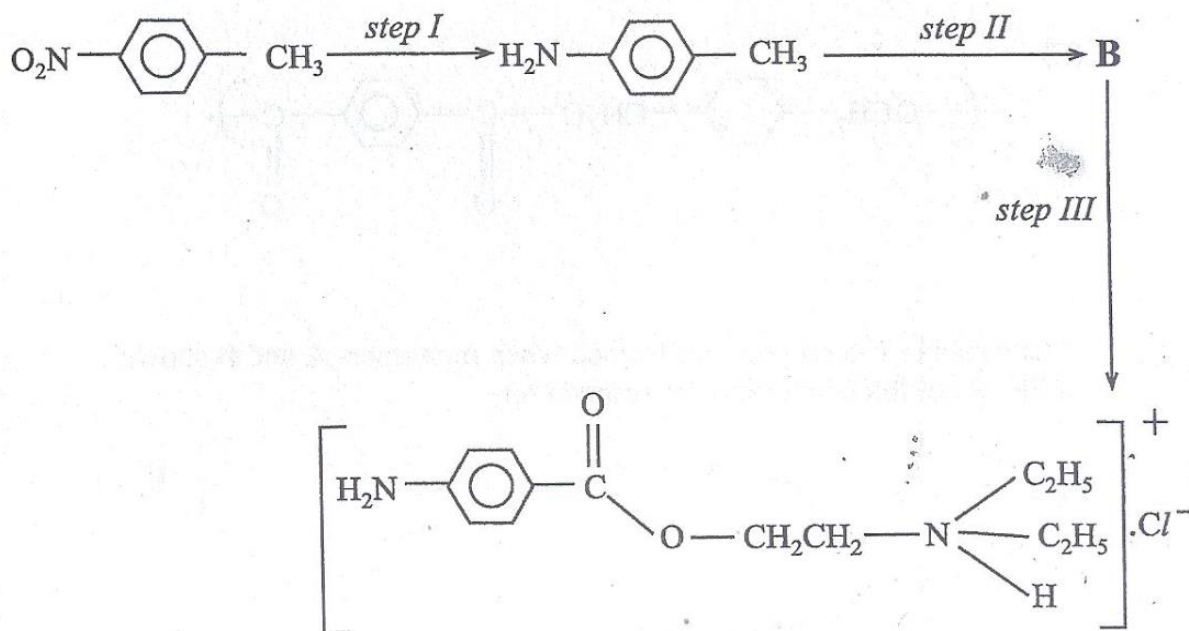


Fig. 2.2

- (i) State the reagents and conditions used in each step.
- (ii) Suggest a structural formula for B.
- (iii) Name the type of reactions which occur in *step I, II* and *III*. [10]
- [Total: 12]

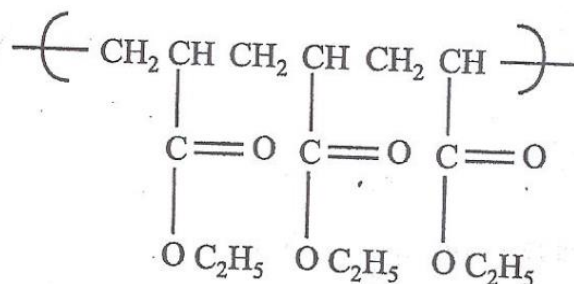
(a) Define the terms

1. *addition polymerisation* and
2. *condensation polymerisation*.

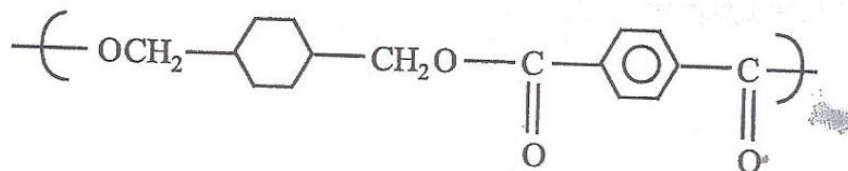
[2]

(b) Draw the structures of the monomers which are used to form the following sections of polymers and name the type of polymerisation involved in their formation.

(i)



(ii)



[5]

(c) Nitrile rubber is a co-polymer formed when monomers A and B shown in Fig. 3 combine in a ratio 2:1 respectively.

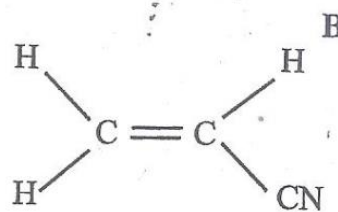
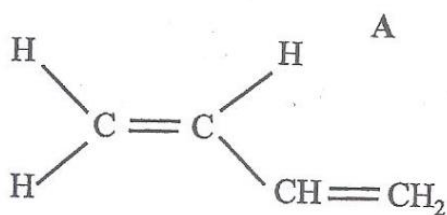


Fig. 3

- (i) Name the monomers A and B.
- (ii) Give a structure showing one repeat unit of the synthetic nitrile rubber.

[3]

- (d) Fig. 4 shows a section of the *all cis* configuration of natural rubber.

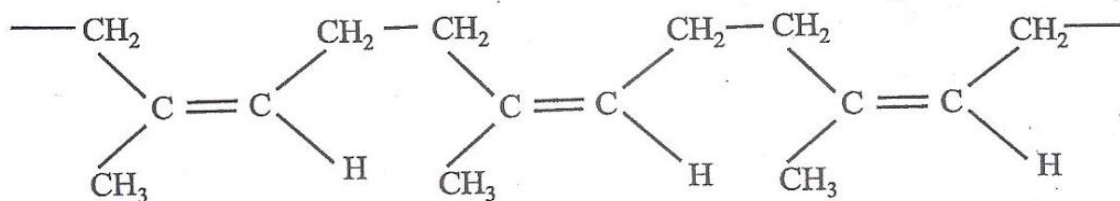


Fig. 4

- (i) Draw the structural formula of the *all trans* configuration of natural rubber.
- (ii) Give the structure of the monomer used to make natural rubber.

[2]

[Total: 12]